Chemistry 3A

Introductory General Chemistry

Experiment 2a

Atomic Masses

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Introduction

In this experiment, you will be calculating the **average atomic mass** of a fictional element called Fresnocitium which has three isotopes that are different colors: **red**, **black**, and **white**.

Facts To Know For This Lab

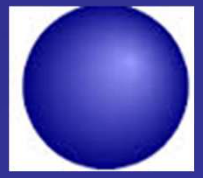
- Matter is composed of **atoms**
- The elements (in the Table) actually describe the properties of these atoms
- Atomic structure shows a composition of **subatomic particles**
  - **Protons**: symbol **p^+**



    - Contained in the **nucleus** of the atom
    - Has a positive charge (+1)
    - Has a mass of **1 amu**, slightly smaller than the neutron, 1800 times larger than the electron
    - Number of protons (atomic number, Z) defines identity of element, its name and symbol
  - **Electrons**: symbol **e^-**



    - Exist in **orbitals** outside of the atomic nucleus
    - Has a negative charge (–1)
    - Has a mass of **5.5×10^{-4} amu**, slightly smaller than the neutron
  - **Neutrons**: symbol **n^0**



    - Exist in nucleus alongside protons
    - Has a zero electrostatic charge
    - Has a mass of about **1 amu**, slightly larger than protons

Background

The element Lithium describes its atoms as follows

- All atoms of lithium have 3 protons (in the nucleus)
- Atoms of lithium can also have either **3** or **4** neutrons in the nucleus
- Atoms that vary with the number of neutrons in the nucleus are called **isotopes**
- Isotopes have several representations for elements
  - Name + mass number: lithium-6, lithium-7
  - Symbol with superscripted mass number: $^{6}\mathrm{Li}$. $^{7}\mathrm{Li}$

Background

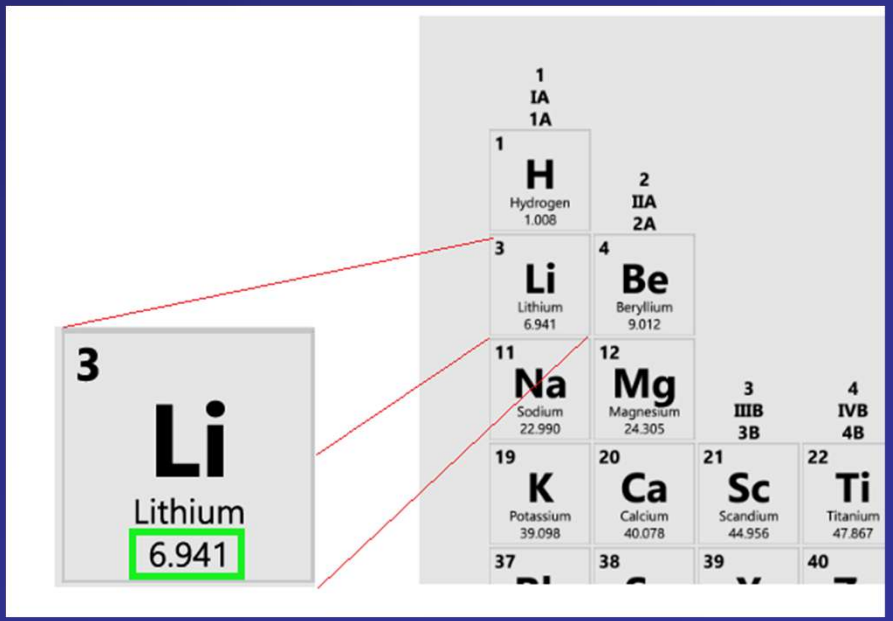
- Lithium thus has two isotopes: lithium-6 and lithium-7. The amu masses and natural abundance are in the table below.

Isotope	Atomic Mass (amu)	Natural Abundance
Lithium-6	6.015	7.59%
Lithium-7	7.016	92.41%

$$atomic\ mass = \sum (\text{relative abundance of isotope})_i \times (\text{isotope mass})_i$$

$$(6.015\ amu)(0.0759) + (7.016\ amu)(0.9241) = \mathbf{6.941\ amu}$$

That calculated number is the one in the Periodic Table for that number: the atomic weight



Equipment You Will Use



electronic balance

Procedure

Look at the Chem 3A lab manual (printed or online)

You are using the balances, so record quantities exactly as they should be recorded

Partners make their own measurements or measure together to verify the masses

Double-check masses when possible. That is repeat the measure a 2nd time: the value should be close. If not, consider instrument issues or not being careful in measuring.

Clean Up

- Make sure the colored beads (“atoms”) are stored in the stock container

Example Data Analysis

DATA

1. Mass of beaker (empty): 68.1410 g

	Trial 1	Trial 2
Mass of beaker + "Nature's Mix" of Fresnocitium	111.3946 g	117.7580 g
Mass of beaker + Red Atoms	81.1362 g	81.4226 g
Mass of beaker + Black Atoms	90.0048 g	92.0999 g
Mass of beaker + White Atoms	76.5318 g	80.0091 g
Number of Red Atoms	43	45
Number of Black Atoms	43	48
Number of White Atoms	17	24

Example Data Analysis

CALCULATIONS

Show your work neatly, with units, where applicable.

2. A. Mass of "Nature's Mix" of Fresnocitium:

TRIAL 1	TRIAL 2
111.3946 - 68.1410 = 43.2536 g	117.7580 - 68.1410 = 49.6170 g

B. Total Number of Atoms:

TRIAL 1	TRIAL 2
43 + 43 + 17 = 103	45 + 48 + 24 = 117

writing "103 atoms" is optional but helpful

Exp 2a (Intro) *Name:*

C. Average Mass per Atom in Nature's Mix:

TRIAL 1	TRIAL 2
43.2536 g / 103 "atoms" = 0.4199 g / "atom"	49.6170 g / 117 "atoms" = 0.4241 g/"atom"

Example Data Analysis

3. A. Complete the table below for Trial 1:

	TRIAL 1	
Isotope Color	Percent Abundance of Isotope (%):	Average Mass per atom of isotope (g):
Red	43/103x100%= 41.75%	(81.1362 g - 68.1410 g) /43 "atoms"= 0.3022 g/"atom"
Black	43/103x100%= 41.75%	(90.0084 g - 68.1410 g) /43 "atoms"= 0.5085 g/"atom"
White	17/103x100%= 16.50%	(76.5318 g - 68.1410 g) /17 "atoms"= 0.4936 g/"atom"

Trial 1: Average Atomic Mass based on equation:

(0.4175 x 0.3022 g) + (0.4175 x 0.5085 g) + (0.1640 x 0.4936 g) =

0.4199 g

B. Complete the table below for Trial 2:

	TRIAL 2	
Isotope Color	Percent Abundance of Isotope (%):	Average Mass per atom of isotope (g):
Red	45/117 x 100% = 38.46%	(81.4226 g - 68.1410 g) /45 "atoms"= 0.2951 g/"atom"
Black	48/117 x 100% = 41.03%	(92.0999 g - 68.1410 g) /48 "atoms"= 0.4991 g/"atom"
White	24/117x100%= 20.51%	(80.0091 g - 68.1410 g) /24 "atoms"= 0.4945 g/"atom"

Trial 2: Average Atomic Mass based on equation:

(0.3846 x 0.2951 g) + (0.4139 x 0.4991 g) + (0.2051 x 0.4945 g) =

0.4197 g

Example Data Analysis

Exp 2a (Intro) *Name:*

4. Compare the "average mass per atom" obtained in Question 2C to the "average atomic mass" obtained in Question 3A for Trial 1. Do these values match? Should they match?

   Write in complete sentences and include both numerical values obtained.

   The average mass per atom from question 2 was 0.4199 g, and the average atomic mass obtained for question 3 was also 0.4199 g. These values match. They should match because both calculations represent the same average mass of the atoms in the sample and is calculated using 2 different methods.

5. Are your results from Trial 1 similar to the results found in Trial 2 for Question 3? If not, why not? (Refer back to the tables in Question 3 to determine what was different between trials)

   Write in complete sentences and include numerical values obtained.

   Yes, the results from Trial 1 are similar to the results from Trial 2. In Trial 1 the average atomic mass was 0.4199 g, and in Trial 2 it was 0.4197 g. These values are very close. The small difference is due to slight changes in the percent abundances and the masses of each isotope between the two trials, which causes small measurement differences.

6. If you used a much larger sample of fresnocitium, would it make a substantial difference to your calculated average atomic mass? Why or why not?

   Write in complete sentences.

actual student answers. looking for reasonable understanding of concept. answers will vary.

   If a much larger sample of fresnocitium were used, it would not make a very big difference in the calculated average atomic mass. A larger sample would make the percent abundance more consistent and reduce random variation, so the average atomic mass number would be more accurate and closer to the true value. Therefore, the results would not be drastically different.